QP CODE: UGII/MOD/07/10/26 Roll no.....

MAR AUGUSTHINOSE COLLEGE RAMAPURAM

MGU-BCA(H) MODEL EXAMINATION, OCTOBER 2026

MG3CCRBCA200- QUANTITATIVE TECHNIQUES

Time: 2 hr **Third semester** **Max.Mark: 70**

Remember(K), Understand(U), Apply(A), Analyze(An), Evaluate(E)

Part A

*Answer any **five** questions. Each question carries 2 marks*

1. Define population and sample? [K] [1]
2. Define statistics? [K] [1]
3. Define ogive? [K] [2]
4. Explain partion values? [U] [2]
5. What are the properties of a good measure of dispersion? [U] [3]
6. Define Scatter Diagram? [K] [3]
7. Define conditional probability? [U][4]
8. State addition theorem of probability. [U] [4]

(5 X 2 = 10)

Part B

*Answer any **five** questions. Each question carries 6 marks*

9. Explain difference between primary data and secondary data. [U] [1]
10. Write a short note on frequency curve and frequency polygon. [U] [1]
11. The mean height of 20 male workers in a factory is 170cm and the mean height of 30 female workers in the same factory is 150cm. Find the combined mean height of 50 workers in the factory. [An][2]
12. Find the mean

Marks	10-20	20-30	30-40	40-50	50-60
No.of students	15	10	12	8	6

[U][2]

13. Find standard deviation.

Wages	0-10	10-20	20-30	30-40	40-50
No.of workers	5	8	15	16	6

[An][2]

14. In a fancy-dress competition, 2 judges awarded the following ranks to 8 participants. Calculate coefficient of rank correlation:

Judge A	8	7	6	3	2	1	5	4
Judge B	7	5	4	1	3	2	6	8

[A][3]

15. Identify the 2 regression equations $5x-6y+90=0$ and $15x-8y-130=0$. [A][3]
16. A university has to select an examiner from a list of 50 persons. 20 of them are women and 30 of them are men; 10 of them know Hindi and 40 do not; 15 of them are teachers and the remaining 35 are not. What is the probability that the university will select a Hindi knowing woman teacher? [A][4]

(5 X 6= 30)

Part C

Answer any two questions. Each question carries 15 marks

17. Explain functions and limitations of statistics. [An][1]
18. Prices of a particular commodity in 5 years in 2 cities A and B are given below. Find the city which has more stable prices.

City A	20	22	19	23	16
City B	10	20	18	12	15

[An] [2]

19. The following data shows maximum and minimum temperature on a certain data at 10 important cities.

Max.temp.	29	23	25	15	27	29	24	31	32	35
Min.temp.	8	3	7	5	8	19	10	7	5	8

Fit the regression line of X on Y and Y on X. [E][3]

20. (a) State Baye's Theorem.
    (b) There are 2 bags, one containing 5 white and 4 black balls and the other containing 6 white and 5 black balls. One bag is selected and one ball is drawn. If it is white, what is the probability that the bag selected is the first? [A][4]

(2 X 15 = 30)

$P(B|W) = 1/2$

$P(B|E) = 15/50$

$P(B|NT) = 35/50$

$P(B|\bar{T}) = 10/50$

$P(B|NH) = 40/50$

$P(W) = 20/50$

$P(\bar{W}) = 30/50$

0.50